Tan Ngo

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EDUCATION

Hanoi University of Science and Technology (HUST) Aug 2021 – Apr 2025

B.Sc. in Computer Science, Hanoi, Vietnam (GPA: 3.69/4.0)

- Graduated early in **3.5 years** and recognized as one of 4 outstanding graduating students (Director's Certificate of Merit).
- Selected to deliver the graduation speech on behalf of all graduating students.

Lam Son High School for the Gifted — *Specialized Mathematics Program, Vietnam* Aug 2018 – May 2021

- **Vietnam National Mathematics Competition, Second Prize — TST List** (2021) — Ranked 11th nationally and selected for the Team Selection Test, the second round for selecting Vietnam's International Mathematical Olympiad team.
- **Vietnam National Mathematics Competition, Consolation Prize** (2020) — National award in Vietnam's highest-level mathematics competition for gifted high school students.
- **North Central Region Mathematics Olympiad, First Prize** (2019) — Among specialized high schools in the North Central Region of Vietnam.

RESEARCH INTEREST

I began as a generative AI engineer focused on **parallel inference and optimization** for generative AI, running experiments to find the fastest way to serve diffusion and language models in production. As an **AI Research Resident** on the Computer Vision Team at **Qualcomm AI Research Vietnam**, under the supervision of Phong Ha Nguyen (Senior Research Scientist, Qualcomm), Binh-Son Hua (Assistant Professor, Trinity College Dublin), and Hung H. Bui (VP, Qualcomm; ex-DeepMind), I worked on distillation of image-generation models and controllable video diffusion models, including the video models' application to generating data for robot-learning training. Currently, I am focused on **efficient ML**, **parallel and distributed computing**, and **ML systems**, and I am passionate about pursuing this as my PhD research direction.

PUBLICATIONS

Track the Noise, Move the World: 3D-Grounded Motion-Consistent Noise for Controllable Video Generation

- Long Van Vu*, **Tan Van Ngo***, Animesh Karnewar, Amir Habibian, Binh-Son Hua, Hung H. Bui, Minh Hoai, and Phong Ha Nguyen. *Under review at NeurIPS 2026.* · ***Co-first author** (equal contribution). · Link

RESEARCH AND WORK EXPERIENCE

AI Research Resident, Qualcomm AI Research Vietnam Jun 2025 – Present

- **Unifying Camera and Motion Control for Video Generation** (Jan 2026 – Present)
  - Developing a method to jointly control camera movement and object motion in 3D space in a video diffusion model, so a user can specify both the trajectory of the scene and the movement of subjects within it — a capability prior work typically handles only one of at a time.
  - Introducing depth estimation as an explicit 3D signal so the model reasons about scene geometry, rather than relying on 2D cues alone, to improve motion consistency across frames.
  - Scaling training, inference, and evaluation across **multi-node GPU clusters (up to 300 GPUs)** on RUNAI, working on data preparation, synchronization, throughput, and training stability.
  - Fixed a model-loading default in **DiffSynth-Studio** that made all 96 GPUs (12 nodes × 8) queue behind one another at startup, re-downloading weights already on disk: per-node startup **9 min → ∼44 s (∼12×)**, GPU start spread within a node **267 s → ∼11 s**, a recurring multi-node crash removed, inference throughput unchanged.
- **Self-Directed Systems Research** (2026 – Present)
  - Favourite courses: Stanford CS336 · Karpathy's series · tinytorch · micrograd · GPU-MODE.
  - Customized **Ray** to make my own research loop **5.4× faster**. Rebuilt my multi-metric video-evaluation pipeline as a pool of workers pulling from a shared queue with a from-scratch tree-reduction, replacing a shell launcher that split work unevenly and silently skipped data — **5.4× faster on 8 GPUs** with identical scores.

- **Ray** wasn't slow — its defaults were, and fixing them sped up my own pipeline. The data transport ran at full hardware speed, but out of the box each worker got a single CPU thread and every step round-tripped through the driver — so CPU-bound work ran **4–9× slower** and coordination cost **∼30× more time than computing**, with nothing crashing or warning.
- A one-line config fix gave **6.7×** on the same hardware: on an 8× H100 node (198 CPU cores), one worker per GPU meant **Ray**'s `num_cpus` acted as a scheduling token, not thread affinity — so CPU-bound metrics (LPIPS, SSIM, PSNR) ran single-threaded and serial. Setting `torch.set_num_threads(198//8)` to split the cores evenly across workers cut the stage **397.3 s → 59.7 s (6.7×)**.
- Found that adding GPUs can make a pipeline slower: my evaluation ran fastest on **2 of 8 GPUs** and the depth/pose annotation on **4 of 8**, because the container's CPU budget is capped (198 cores) and is split among workers — past a point each extra GPU only waits for CPU.

– **Few-Step Text-to-Multiview Diffusion Model** (Jun 2025 – Dec 2025)
- Distilled a multi-step MVDream text-to-multiview model into a **few-step** student, sharply reducing the number of sampling steps while preserving multiview consistency, for faster interactive and on-device use.
- Raised generation from 256×256 to 512×512 per view — 4× the pixels, above the resolution the teacher itself produced — and improved output quality through training-recipe changes, data refinement, and model optimization.
- **Quantized** the few-step model and supported the applied team in deploying it on Samsung S24 phones; the result was shown as a live demo at **Qualcomm's NeurIPS 2025** booth.

Generative AI Engineer, ZenAI Aug 2024 – Jun 2025

– **Diffusion Image-Generation Acceleration (FLUX, SDXL)** (Aug 2024 – Jun 2025)
- **Sped up** image generation for FLUX and SDXL diffusion models in production:
  - **FLUX.1-schnell (8-step):** context parallelism (2× H100) + `torch.compile` fusion + first-block caching cut inference **2.02 s → 0.86 s (≈2.35× faster)** at ≥95% image quality.
  - **SDXL Turbo (8-step):** CFG parallelism (2× H100) + `torch.compile` cut inference **1.26 s → 0.72 s (≈1.75× faster)** at ≥99% image quality.
- **Fixed a bug in `torch.compile`** that was silently blocking these speedups (torch 2.3.1): traced an opaque compiler error to a single operation whose index expression couldn't be built, then added a fallback so that operation degrades gracefully instead of killing the entire compilation — restoring the compiled path.
- **Dynamic-shape problem** — found the limit of the technique itself: `torch.compile`'s speedup holds only for a small set of distinct input shapes (∼16); beyond **∼100 shapes** it silently falls back to uncompiled speed as each new shape triggers a recompilation. The fix: serve a small bucketed set of shapes rather than compiling per shape.

– **LLM Serving & Inference Optimization (vLLM, SGLang)** (Aug 2024 – Jun 2025)
- Served and optimized large language models with the vLLM and SGLang frameworks for production workloads.
- Applied **quantization**, **KV-cache** optimization, and **parallel** GPU execution to cut **latency** and raise **throughput** under production serving loads.
- Reduced CPU–GPU communication overhead to improve end-to-end serving **efficiency**.

Research Assistant, Computer Vision Lab, BKAI, HUST Oct 2023 – Jan 2025

– **Continual Learning for Object Detection with Generative Data Replay** (Sep 2024 – Jan 2025)
- Applied continual learning to object detection to mitigate catastrophic forgetting as new tasks/classes arrive sequentially, instead of retraining on all data.
- Used diffusion-generated replay data to stand in for past-task samples, improving knowledge retention and training stability without storing the original datasets.

– **Vision Transformers for Medical Endoscopy Image Segmentation** (Oct 2023 – Aug 2024)
- Researched Vision Transformer–based architectures for segmentation of medical endoscopy images, using Masked Autoencoder (MAE) self-supervised pretraining on unlabeled data to improve downstream tasks.

HONORS AND AWARDS

HUST Wings Scholarship — Awarded for six consecutive semesters. 2022–2024

HUST Academic Encouragement Scholarship — Computer Science department, <3% selected. 2021–2022

TECHNICAL SKILLS

Languages: Python, C/C++
Distributed & parallel: multi-GPU / multi-node training and inference (up to 300 GPUs), NCCL, CPU/GPU workload balancing, Ray, SLURM
Efficiency & compilers: `torch.compile`, quantization, KV-cache, distillation, vLLM, SGLang
Frameworks & libraries: PyTorch, Diffusers, ComfyUI, NumPy, OpenCV

REFERENCES

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